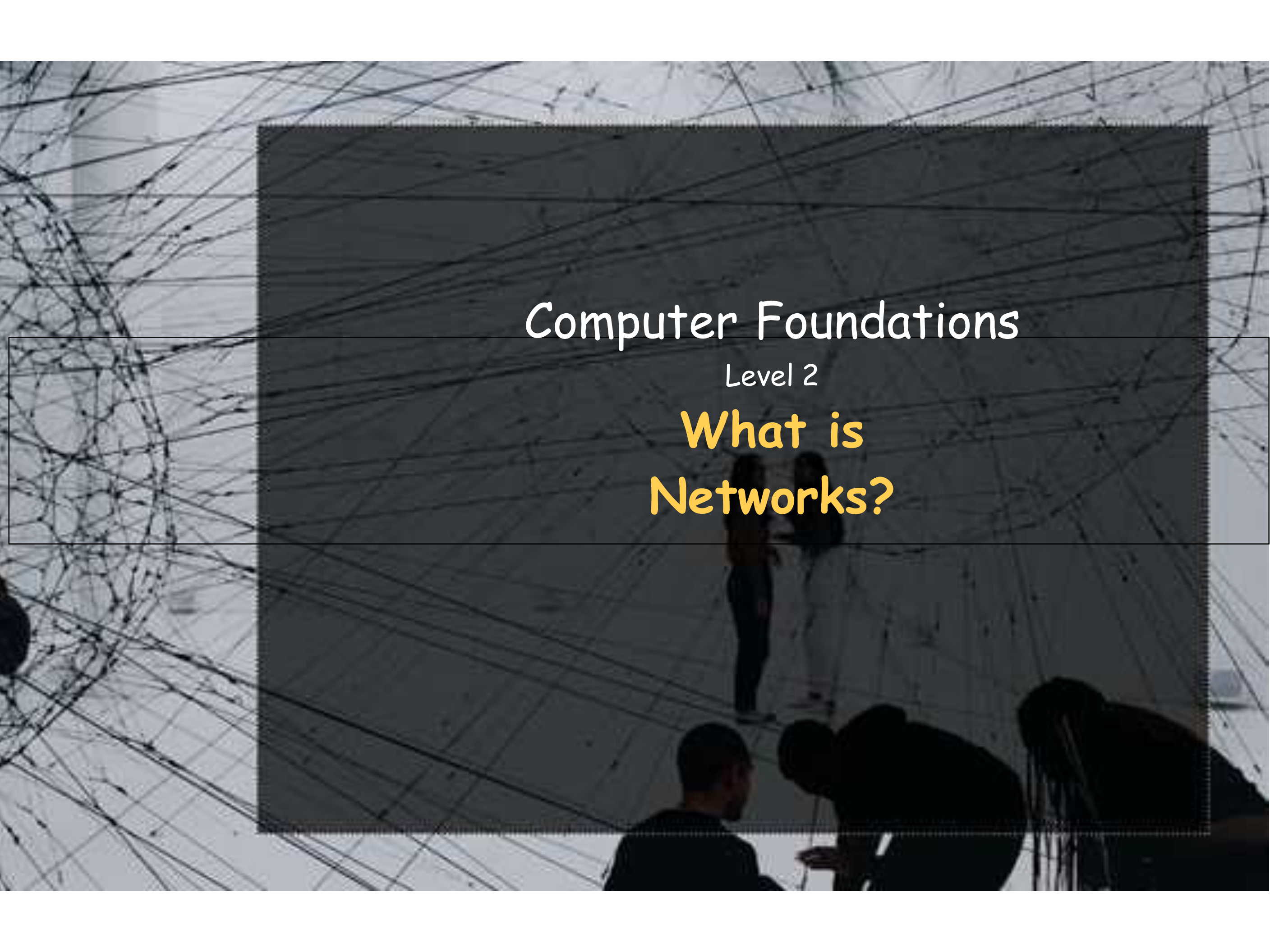




Computer Foundations Level 2

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing the entire frame, resembling a web or a dense circuit board. In the lower portion of the image, there are dark silhouettes of several people. Two individuals are standing in the center, facing each other, while others are seated or crouching in the foreground, suggesting a collaborative work environment or a meeting. The overall color palette is dark and monochromatic, with the text providing the primary source of color contrast.

Computer Foundations

Level 2

**What is
Networks?**

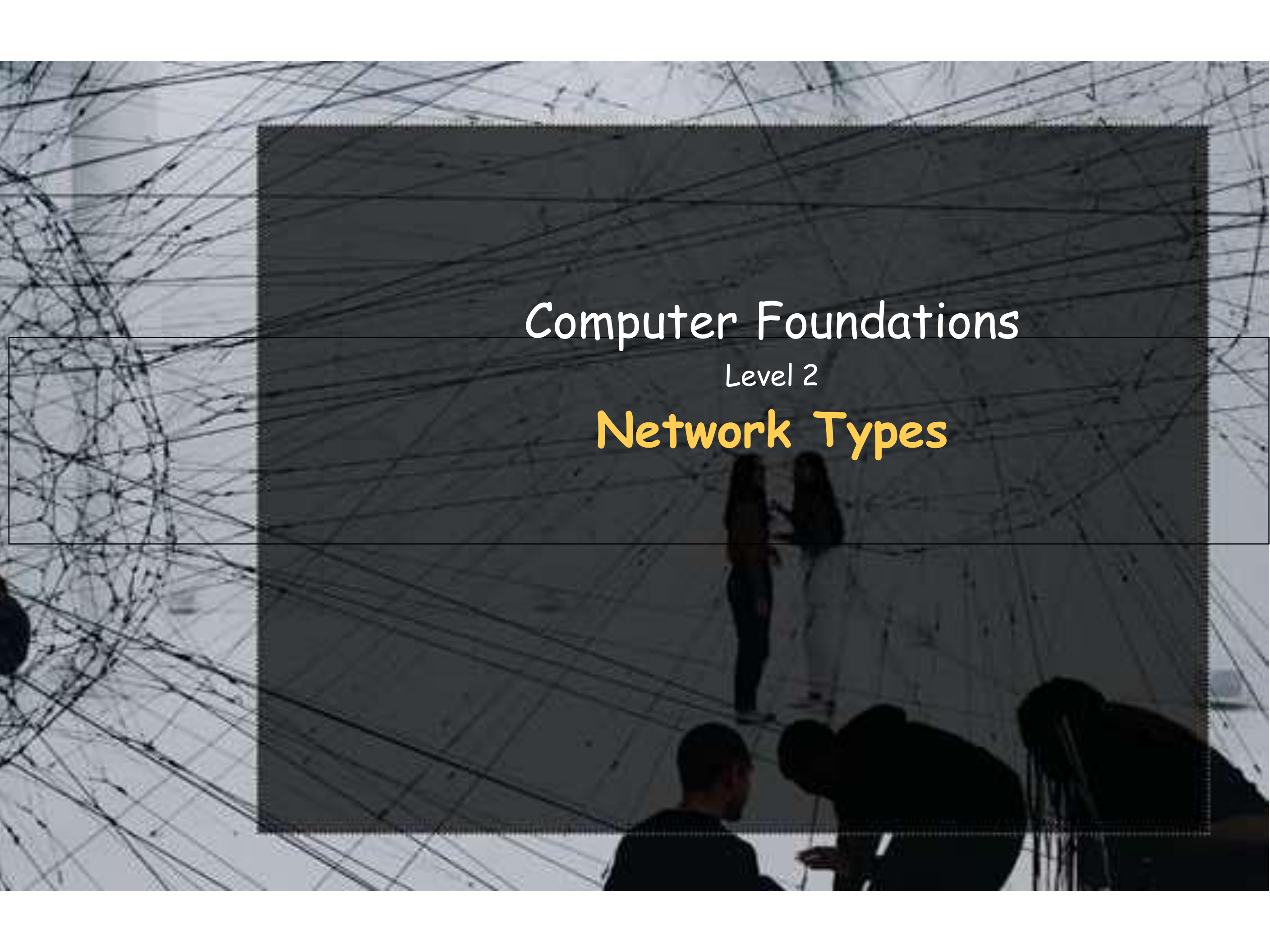
What is Network?

Network: is a group of connected computer and devices.

- For example, the computers, phones, and other devices that your family owns are all part of your home computer network.
- When you get a new tablet and enter in the Wi-Fi password, you're joining that tablet to your home computer network.

Why do we need Network?

- Devices in the same network can easily communicate and share resources with each other.
- For example : you can control your TV with your mobile, or you can stream videos from your mobile to your TV.



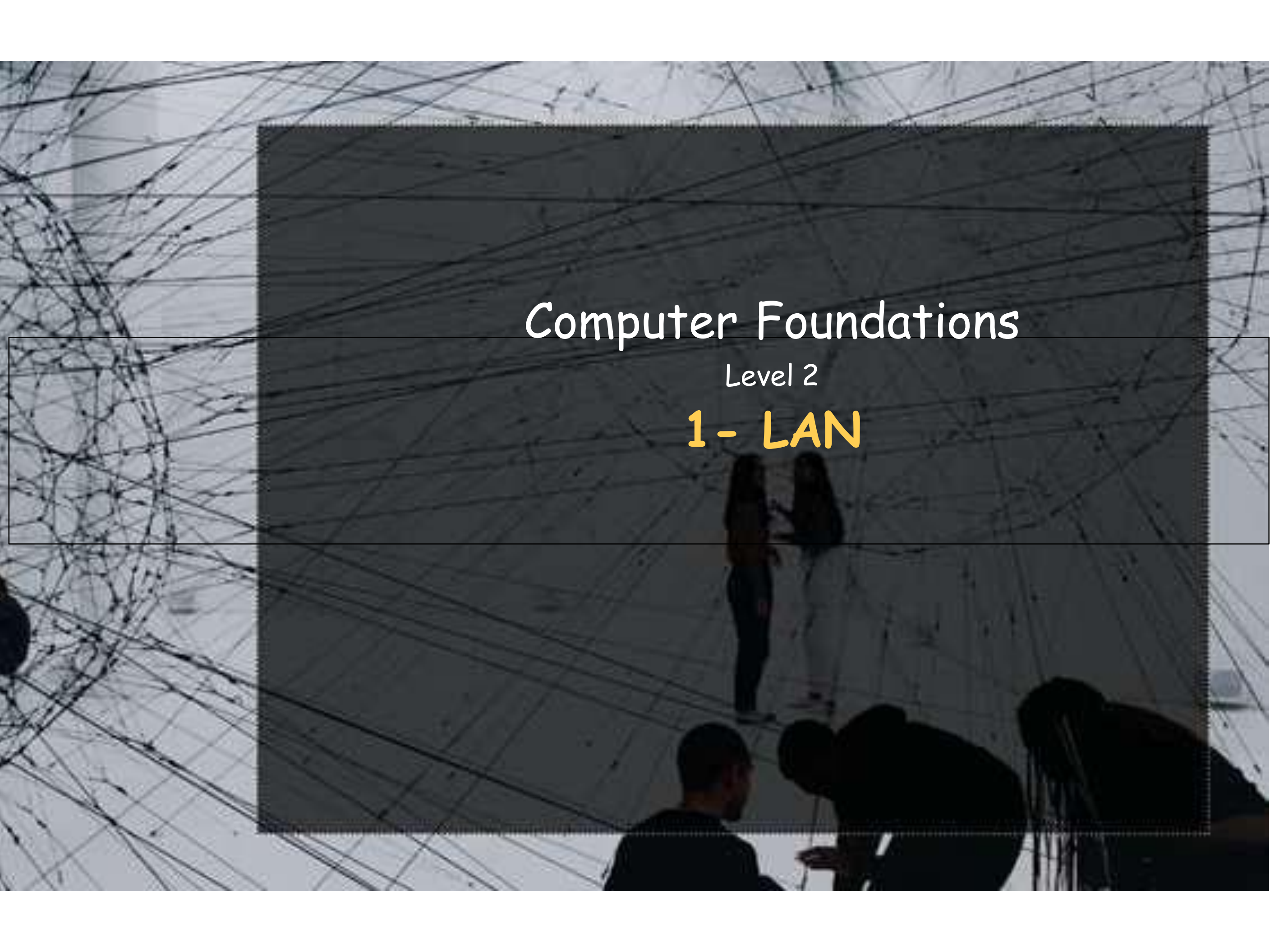
Computer Foundations

Level 2

Network Types

Network Types





Computer Foundations

Level 2

1- LAN

What is LAN?

LAN: Local Area Network that has a group of connected devices.

- LAN is a type of network made up of nearby computers.
- A LAN can have anywhere from a few devices (like a home network) to hundreds of devices (like a school or hospital)
- Large organizations like colleges can have multiple LANs - maybe one per building.

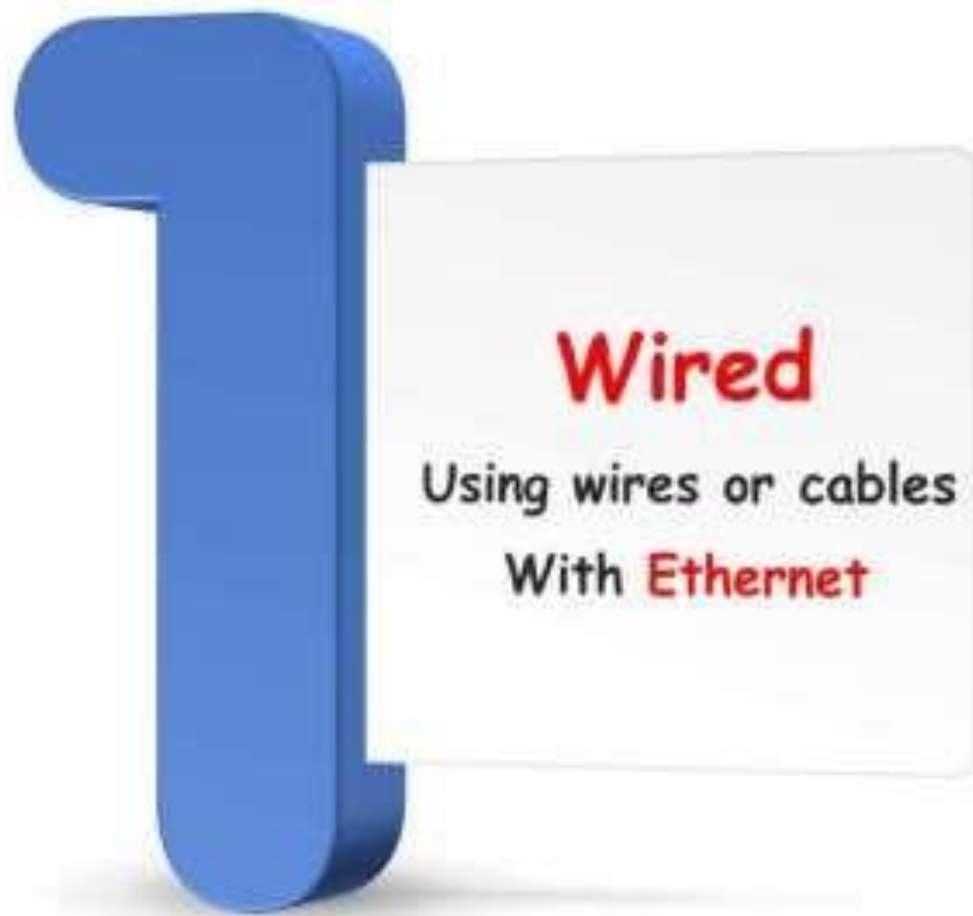


Computer Foundations

Level 2

Communicating on LAN

Connection Types





Computer Foundations

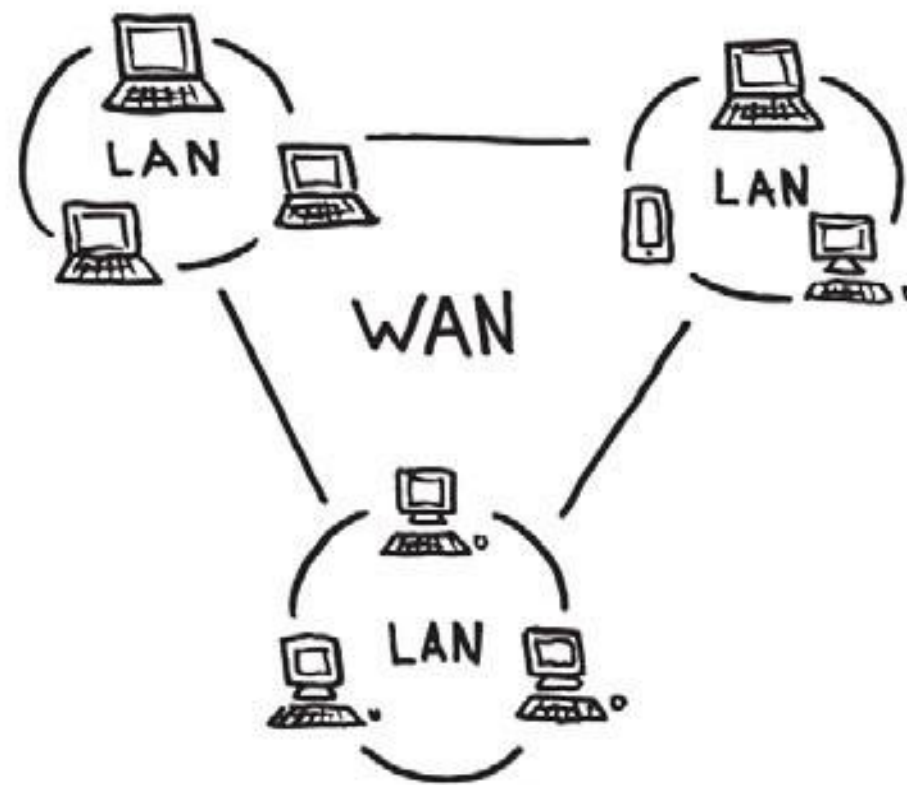
Level 2

2- WAN

What is WAN?

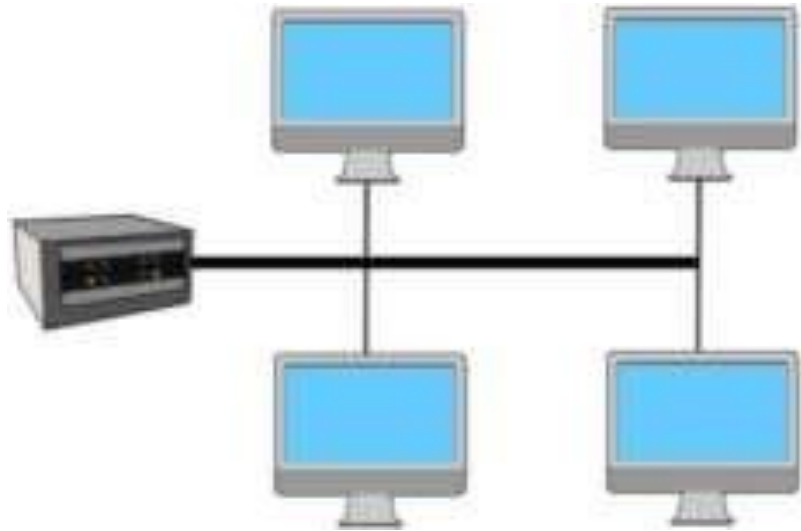
WAN: Wide Area Network.

- WAN is made of connected LANs that are far from each other (like in different parts of the country or world).
- WANs connect thousands or even millions of devices.



Communication Ways

Switch



Ethernet

Rules that computers follow for communicating on a wired network



Router



Wireless

Rules that computers follow for communicating through a radio signals

Those Rules are called:

Protocol

Protocol: is a standard set of rules for how devices communicate through wires or wireless

Wi-Fi is a popular way to connect wirelessly to a network.

Wi-Fi: Wireless Fidelity

Homework



Answer the following questions:

- What is network?
- Why do we need network?
- What is LAN vs WAN?
- How computers communicate together?
- What is Ethernet?
- What is Wireless?
- What is Protocol?
- What is Wi-Fi?
- Wi-Fi stands for what?

Thank you 😊